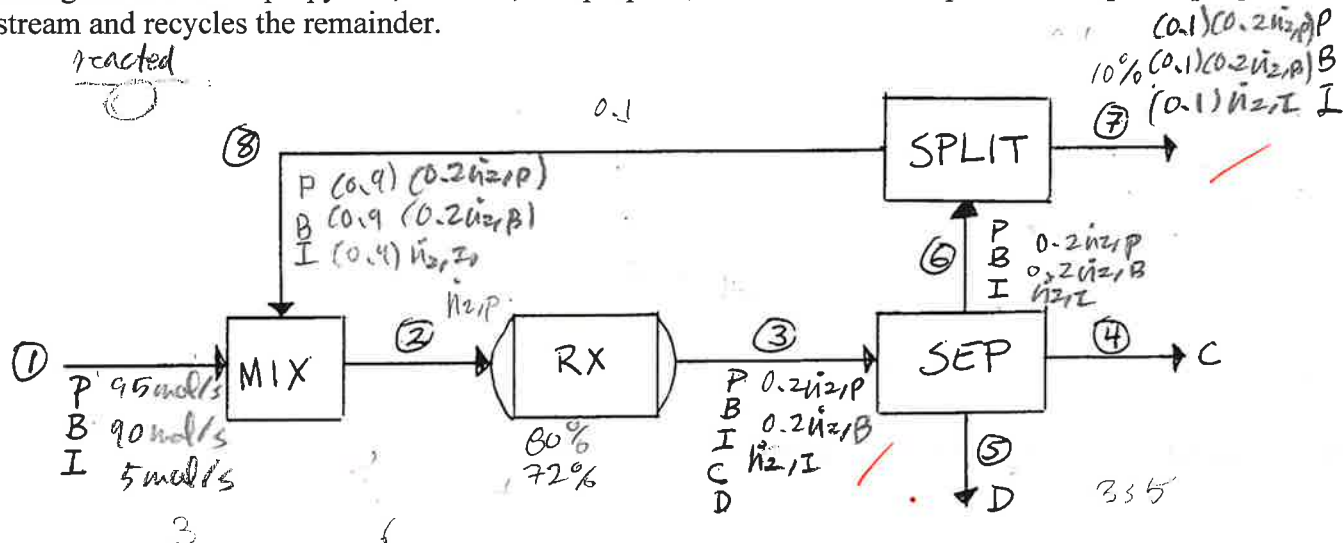


To receive full credit, you must show all of your work.

- (1) (60 pts.) Cumene (C_9H_{12}) is synthesized from propylene (C_3H_6) and benzene (C_6H_6):



The undesired byproduct diisopropylbenzene ($C_{12}H_{18}$) is produced in a second competing reaction. Refer to the block flow diagram below. The fresh feed contains 95 mol/s propylene, 90 mol/s benzene, and 5 mol/s of propane I (C_3H_8) which is inert. This stream is combined in the mixer with the recycle stream. The reactor single-pass conversion is 80% of the propylene and 72% of the benzene. The reactor product goes to a separator which has three output streams: a stream of pure cumene, a stream of pure diisopropylbenzene, and a stream containing the unreacted propylene, benzene, and propane, which is sent to a splitter. The splitter purges 10% of this stream and recycles the remainder.



- (a) Finish labeling the flow diagram above. In the column "Entire Process" (see Table below), use the degree of freedom analysis (see p. 105 of Murphy) to determine if the problem is well posed. Give a brief explanation for each entry. (10 pts.)

	Entire Process	Explanation
4a # stream variables	22 ✓	every variables in every streams
4b # reaction variables	2 ✓	2 reactions
# indep variables	24 ✓	$22 + 2 = 24$
5a # specified flows	3	in 1 stream, P, B, I have flow rates.
5b # spec stream comps and split	2 ✗	80%, 72% of compositions
5c # performance specs	3	2 reactions + 10% purge ✓
5d # material balances	16 ✓	number of components in each system.
# indep eqns	24 ✓	$3 + 2 + 3 + 16 = 24$
DoF	0	$24 - 24 = 0$

Problem 2 continued

- (b) Calculate the fractional yield for cumene for the following percent excesses in benzene: 0%, 100%, and 400%. Be careful to avoid roundoff error. (15 pts.)

$$0\% = \frac{4\bar{y}_1}{0.0005(10 - \bar{y}_1)} \quad \bar{y}_1 = 0$$

$$1 = \frac{4\bar{y}_1}{0.0005(10 - \bar{y}_1)}$$

$$4\bar{y}_1 = 0.0005(10 - \bar{y}_1)$$

$$\bar{y}_1 = 1.25 \times 10^{-3} \text{ mol/s}$$

$$4 = \frac{4\bar{y}_1}{0.0005(10 - \bar{y}_1)}$$

$$4\bar{y}_1 = 4(0.0005(10 - \bar{y}_1))$$

$$\bar{y}_1 = 5.00 \times 10^{-3} \text{ mol/s}$$

$$f_{\text{yield}}^A = \frac{n_{2,c}}{n_{1,B}} = \frac{(2\bar{y}_1)}{(10 \text{ mol/s})}$$

% Excess Benzene	0%	100%	400%
Fractional Yield	0	2.5×10^{-4}	1×10^{-3}

- (c) Suggest how the design of the process could be changed to improve the fractional yield of cumene. (5 pts.)

As putting more benzene, the fractional yield is getting increase
Therefore, we need to design the process with the enough benzene

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Problem 1 continued

(b) Write the appropriate balance equations. Calculate the molar flow rates in the process in mol/s. Be sure to define the system for any sets of balance equations you write! Fill in the Table below with your results. (45 pts.)

describe molar rates in streams in terms of $\dot{n}_{2,P}$ by using conversion 80% & 72%, and purges 10%.

$$\dot{n}_{3,P} = 0.2 \dot{n}_{2,P} \quad \dot{n}_{7,P} = (0.1)(0.2 \dot{n}_{2,P})$$

$$\dot{n}_{6,P} = 0.2 \dot{n}_{2,P} \quad \dot{n}_{8,P} = (0.9)(0.2 \dot{n}_{2,P})$$

$$\begin{array}{r} 115.85 \\ \times 2.26 \\ \hline 261.78 \\ \hline 254.82 \end{array}$$

$$\dot{n}_{3,B} = 0.28 \dot{n}_{2,B} \quad \dot{n}_{7,B} = (0.1)(0.28 \dot{n}_{2,B})$$

$$\dot{n}_{6,B} = 0.28 \dot{n}_{2,B} \quad \dot{n}_{8,B} = (0.9)(0.28 \dot{n}_{2,B})$$

Balance in mixer:

$$\dot{n}_{1,P} + \dot{n}_{8,P} = \dot{n}_{2,P} \Rightarrow 95 \text{ mol/s} = \dot{n}_{2,P} - 0.9(0.2 \dot{n}_{2,P})$$

$$\dot{n}_{2,P} = 95 \text{ mol/s} / (1 - 0.9 \times 0.2) = 115.85 \text{ mol/s}$$

$$\dot{n}_{1,B} + \dot{n}_{8,B} = \dot{n}_{2,B} \Rightarrow 90 \text{ mol/s} = \dot{n}_{2,B} - 0.9(0.28 \dot{n}_{2,B})$$

$$\dot{n}_{2,B} = 90 \text{ mol/s} / (1 - 0.9 \times 0.28) = 120.32 \text{ mol/s}$$

Inert: $5 \text{ mol/s} + (0.9) \dot{n}_{2,I} = \dot{n}_{2,I} \Rightarrow \dot{n}_{2,I} = 50 \text{ mol/s}$

Balance in reaction: $P + B \rightarrow C$ (ξ_1) $P + C \rightarrow D$ (ξ_2)

$$P: \dot{n}_{2,P} - \xi_1 - \xi_2 = \dot{n}_{3,P} = 0.2 \dot{n}_{2,P} \quad (1)$$

$$B: \dot{n}_{2,B} - \xi_1 = \dot{n}_{3,B} \quad (2)$$

$$C: \xi_1 - \xi_2 = \dot{n}_{3,C} \quad (3)$$

$$D: \xi_2 = \dot{n}_{3,D} \quad (4)$$

Stream Summary Table: All flows in mol/s

↓/Stream→	1	2	3	4	5	6	7	8
P	95	115.85	23.17	-----	-----	23.17	2.32	20.85
B	90	120.32	33.69	-----	-----	33.69	3.37	30.32
I	5	50	50	-----	-----	50	5	45
C	-----	-----	80.58	80.58	-----	-----	-----	-----
D	-----	-----	6.05	-----	6.05	-----	-----	-----

(c) Calculate the overall fractional conversions of propylene, the fractional selectivity, and the fractional yield (5 pts.)

$$f_{\text{conversion}}^P = \frac{115.85 \text{ mol/s} - 23.17 \text{ mol/s}}{95 \text{ mol/s}} = 0.975$$

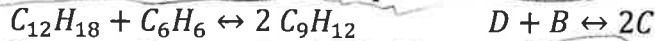
$$f_{\text{selectivity}}^P = \frac{80.58 \text{ mol/s}}{115.85 \text{ mol/s} - 23.17 \text{ mol/s}} = 0.869$$

$$f_{\text{yield}}^P = \frac{80.58 \text{ mol/s}}{95 \text{ mol/s}} = 0.848$$

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- (2) (40 pts.) In the production of cumene, diisopropylbenzene is produced as a byproduct. To avoid wasting this material it can be converted to cumene in an equilibrium reaction:

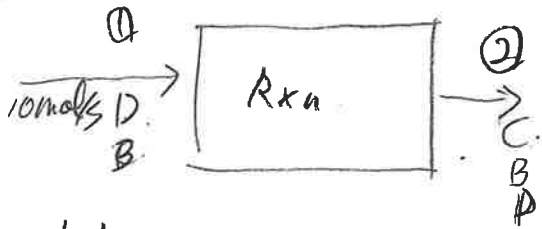


A stream of 10 mol/s diisopropylbenzene is mixed with excess benzene and fed to an equilibrium reactor. The relation between the equilibrium constant and the mole fractions in the product stream is:

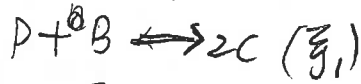
$$K = \frac{x_C^2}{x_D x_B}$$

where under the reactor conditions $K = 0.0005$, which is a small value favoring the reactants.

- (a) Write the appropriate species balance equations for this reaction. Combine these results with the equilibrium constant relation to derive a general expression for the percent excess benzene. (20 pts.)



reaction balance



$$\dot{n}_{1,D} - \xi_1 = \dot{n}_{2,D}$$

$$\dot{n}_{1,B} - \xi_1 = \dot{n}_{2,B}$$

$$2\xi_1 = \dot{n}_{2,C}$$

$$x_C = \frac{\dot{n}_{2,C}}{\dot{n}_{2,C} + \dot{n}_{2,D} + \dot{n}_{2,B}}$$

$$x_D = \frac{\dot{n}_{2,D}}{\dot{n}_{2,C} + \dot{n}_{2,D} + \dot{n}_{2,B}}$$

$$x_B = \frac{\dot{n}_{2,B}}{\dot{n}_{2,C} + \dot{n}_{2,D} + \dot{n}_{2,B}}$$

$$K = \frac{(\dot{n}_{2,D})^2}{(\dot{n}_{2,C} + \dot{n}_{2,D} + \dot{n}_{2,B})^2} = \frac{(2\xi_1)^2}{(\dot{n}_{1,B} - \xi_1)(10 - \xi_1)} = 0.0005$$

$$0.0005(\dot{n}_{1,B} - \xi_1)(10 - \xi_1) = (2\xi_1)^2$$

$$\dot{n}_{1,B} - \xi_1 = \frac{4\xi_1^2}{0.0005(10 - \xi_1)}$$

$$\dot{n}_{1,B} = \frac{4\xi_1^2}{0.0005(10 - \xi_1)} + \xi_1$$

$$\text{EXCESS \% of B} = \left(\frac{\dot{n}_{1,B}}{\xi_1} - 1 \right) \left(1 + \frac{4\xi_1}{0.0005(10 - \xi_1)} - 1 \right) = \frac{4\xi_1}{0.0005(10 - \xi_1)}$$

$$\begin{aligned} \text{eqn (2)}: \quad \dot{\bar{y}}_1 &= \dot{n}_{2,B} - \dot{n}_{3,B} \\ &= (1 - 0.28) \dot{n}_{2,B} = 0.72 \times 120.32 \text{ mol/s} = 86.63 \text{ mol/s} \end{aligned}$$

$$\begin{aligned} \text{eqn (1)}: \quad \dot{\bar{y}}_2 &= \dot{n}_{2,P} - \dot{\bar{y}}_1 - \dot{n}_{3,P} \\ &= (1 - 0.2) \dot{n}_{2,P} - 86.63 \text{ mol/s} = 0.8 \times 115.85 \text{ mol/s} - 86.63 \text{ mol/s} \\ &= 6.05 \text{ mol/s} \end{aligned}$$

$$\text{eqn (3)}: \quad \dot{n}_{3,C} = 86.63 \text{ mol/s} - 6.05 \text{ mol/s} = 80.58 \text{ mol/s}$$

$$\text{eqn (4)}: \quad \dot{n}_{3,D} = 6.05 \text{ mol/s}$$

$$\dot{n}_{3,P} = 0.2 \cdot \dot{n}_{2,P} = 23.17 \text{ mol/s}$$

$$\dot{n}_{3,B} = 0.28 \dot{n}_{2,B} = 33.69 \text{ mol/s}$$

no reaction in separator $\Rightarrow$ no change ^{like} $\dot{n}_{6,P} = \dot{n}_{3,P}$

balance in split

$$\dot{n}_{7,P} = (0.1) \dot{n}_{6,P} = 0.1 \times 23.17 = 2.32 \text{ mol/s}$$

$$\dot{n}_{7,B} = (0.1) \dot{n}_{6,B} = 0.1 \times 33.69 = 3.37 \text{ mol/s}$$

$$\dot{n}_{7,I} = (0.1) \dot{n}_{6,I} = 0.1 \times 50 = 5 \text{ mol/s}$$

$$\dot{n}_{8,P} = 0.9 \dot{n}_{6,P} = 0.9 \times 23.17 = 20.85 \text{ mol/s}$$

$$\dot{n}_{8,B} = 0.9 \dot{n}_{6,B} = 0.9 \times 33.69 = 30.32 \text{ mol/s}$$

$$\dot{n}_{8,I} = 0.9 \dot{n}_{6,I} = 0.9 \times 50 = 45 \text{ mol/s}$$

